



**AI04SN03 VI**

**LCD Driver IC EC Notice \_Ver.  
3**

GDBU

2024/11/25

# LCD Driver IC EOL And Cover plan



- \* Purpose: Due to the current LCD driver IC supply shortage, ADP will create new cell with Sitronix IC for longevity support.
- \* Change Items: For Sitronix IC and new cell, new product specification need to be changed as below.
  - 1) Electrical Change : page 3 ~ page9
  - 2) Mechanical Change : page 10
  - 3) Optical Change : page 11
  - 4) Packing Change : page 12
  - 5) Label Change : page 13
  - 6) Protection Film Change : page 14
  - 7) Create new p/n to control and change the production site to Suzhou (S06)
    - new p/n : 97D10A07.111
- \* Phase in schedule
  - a. Customer Approval : 2024/12/E
  - c. Phase in : 2025/Q1

# Pin Assignment



Recommended connector : HRS FH28-60S-0.5SH.

| Pin no. | Symbol | I/O | Description                                 | Remark |
|---------|--------|-----|---------------------------------------------|--------|
| 1.      | AGND   | P   | Ground for analog circuit.                  |        |
| 2.      | AVDD   | P   | Analog power supply voltage.                |        |
| 3.      | VDDIO  | P   | Digital interface supply voltage.           |        |
| 4.      | R0     | I   | Red data input (LSB).                       |        |
| 5.      | R1     | I   | Red data input.                             |        |
| 6.      | R2     | I   | Red data input.                             |        |
| 7.      | R3     | I   | Red data input.                             |        |
| 8.      | R4     | I   | Red data input.                             |        |
| 9.      | R5     | I   | Red data input.                             |        |
| 10.     | R6     | I   | Red data input.                             |        |
| 11.     | R7     | I   | Red data input (MSB).                       |        |
| 12.     | G0     | I   | Green data input (LSB).                     |        |
| 13.     | G1     | I   | Green data input.                           |        |
| 14.     | G2     | I   | Green data input.                           |        |
| 15.     | G3     | I   | Green data input.                           |        |
| 16.     | G4     | I   | Green data input.                           |        |
| 17.     | G5     | I   | Green data input.                           |        |
| 18.     | G6     | I   | Green data input.                           |        |
| 19.     | G7     | I   | Green data input (MSB).                     |        |
| 20.     | B0     | I   | Blue data input (LSB).                      |        |
| 21.     | B1     | I   | Blue data input.                            |        |
| 22.     | B2     | I   | Blue data input.                            |        |
| 23.     | B3     | I   | Blue data input.                            |        |
| 24.     | B4     | I   | Blue data input.                            |        |
| 25.     | B5     | I   | Blue data input.                            |        |
| 26.     | B6     | I   | Blue data input.                            |        |
| 27.     | B7     | I   | Blue data input (MSB).                      |        |
| 28.     | DCLK   | I   | Data clock input.                           |        |
| 29.     | DE     | I   | Data enable signal.                         |        |
| 30.     | HSYNC  | I   | Horizontal sync input. (Negative polarity). |        |
| 31.     | VSNC   | I   | Vertical sync input. (Negative polarity).   |        |
| 32.     | SCL    | I   | Serial communication clock input.           |        |
| 33.     | SDA    | I   | Serial communication data input.            |        |
| 34.     | CSB    | I   | Serial communication chip select.           |        |

Change to



| Pin no. | Symbol | I/O | Description                                      | Remark |
|---------|--------|-----|--------------------------------------------------|--------|
| 1       | AGND   | P   | Ground for analog circuit.                       |        |
| 2       | AVDD   | P   | Analog power supply voltage.                     |        |
| 3       | VDDIO  | P   | Digital interface supply voltage.                |        |
| 4       | R0     | I   | Red data input (LSB).                            |        |
| 5       | R1     | I   | Red data input.                                  |        |
| 6       | R2     | I   | Red data input.                                  |        |
| 7       | R3     | I   | Red data input.                                  |        |
| 8       | R4     | I   | Red data input.                                  |        |
| 9       | R5     | I   | Red data input.                                  |        |
| 10      | R6     | I   | Red data input.                                  |        |
| 11      | R7     | I   | Red data input (MSB).                            |        |
| 12      | G0     | I   | Green data input (LSB).                          |        |
| 13      | G1     | I   | Green data input.                                |        |
| 14      | G2     | I   | Green data input.                                |        |
| 15      | G3     | I   | Green data input.                                |        |
| 16      | G4     | I   | Green data input.                                |        |
| 17      | G5     | I   | Green data input.                                |        |
| 18      | G6     | I   | Green data input.                                |        |
| 19      | G7     | I   | Green data input (MSB).                          |        |
| 20      | B0     | I   | Blue data input (LSB).                           |        |
| 21      | B1     | I   | Blue data input.                                 |        |
| 22      | B2     | I   | Blue data input.                                 |        |
| 23      | B3     | I   | Blue data input.                                 |        |
| 24      | B4     | I   | Blue data input.                                 |        |
| 25      | B5     | I   | Blue data input.                                 |        |
| 26      | B6     | I   | Blue data input.                                 |        |
| 27      | B7     | I   | Blue data input (MSB).                           |        |
| 28      | DCLK   | I   | Data clock input.                                |        |
| 29      | DE     | I   | Data enable signal.                              |        |
| 30      | NC     | -   | For test, do not connect (Please leave it open). |        |
| 31      | NC     | -   | For test, do not connect (Please leave it open). |        |
| 32      | NC     | -   | For test, do not connect (Please leave it open). |        |
| 33      | NC     | -   | For test, do not connect (Please leave it open). |        |
| 34      | NC     | -   | For test, do not connect (Please leave it open). |        |

→Change Pin 30~ Pin 34 to NC.

AUO Display+

# Pin Assignment



|     |         |    |                                                                                                                                            |  |
|-----|---------|----|--------------------------------------------------------------------------------------------------------------------------------------------|--|
| 35. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                           |  |
| 36. | VDDIO.  | P. | Digital interface supply voltage.                                                                                                          |  |
| 37. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                           |  |
| 38. | GND.    | P. | Ground for digital circuit.                                                                                                                |  |
| 39. | AGND.   | P. | Ground for analog circuit.                                                                                                                 |  |
| 40. | AVDD.   | P. | Analog power supply voltage.                                                                                                               |  |
| 41. | VCOMin. | I. | For external VCOM DC input.                                                                                                                |  |
| 42. | DITH.   | I. | Dithering setting.<br>DITH = "L" 6bit resolution(LSB last 2 bits of input data truncated).<br>DITH = "H" 8bit resolution(Default setting). |  |
| 43. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                           |  |
| 44. | VCOM.   | O. | connect a capacitor.                                                                                                                       |  |
| 45. | V10.    | I. | Gamma correction voltage reference.                                                                                                        |  |
| 46. | V9.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 47. | V8.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 48. | V7.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 49. | V6.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 50. | V5.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 51. | V4.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 52. | V3.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 53. | V2.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 54. | V1.     | I. | Gamma correction voltage reference.                                                                                                        |  |
| 55. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                           |  |
| 56. | VGH.    | P. | Positive power for TFT.                                                                                                                    |  |
| 57. | VDDIO.  | P. | Digital interface supply voltage.                                                                                                          |  |
| 58. | VGL.    | P. | Negative power supply for Gate driver.                                                                                                     |  |
| 59. | GND.    | P. | Ground for digital circuit.                                                                                                                |  |
| 60. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                           |  |

I: Input; P: Power.

Change to



|     |         |    |                                                                                                                                             |  |
|-----|---------|----|---------------------------------------------------------------------------------------------------------------------------------------------|--|
| 35. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 36. | VDDIO.  | P. | Digital interface supply voltage.                                                                                                           |  |
| 37. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 38. | GND.    | P. | Ground for digital circuit.                                                                                                                 |  |
| 39. | AGND.   | P. | Ground for analog circuit.                                                                                                                  |  |
| 40. | AVDD.   | P. | Analog power supply voltage.                                                                                                                |  |
| 41. | VCOMin. | I. | For external VCOM DC input.                                                                                                                 |  |
| 42. | DITH.   | I. | Dithering setting.<br>DITH = "L" 6bit resolution(LSB last 2 bits of input data truncated) (Default setting).<br>DITH = "H" 8bit resolution. |  |
| 43. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 44. | VCOM.   | O. | connect a capacitor.                                                                                                                        |  |
| 45. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 46. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 47. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 48. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 49. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 50. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 51. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 52. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 53. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 54. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 55. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |
| 56. | VGH.    | P. | Positive power for TFT.                                                                                                                     |  |
| 57. | VDDIO.  | P. | Digital interface supply voltage.                                                                                                           |  |
| 58. | VGL.    | P. | Negative power supply for Gate driver.                                                                                                      |  |
| 59. | GND.    | P. | Ground for digital circuit.                                                                                                                 |  |
| 60. | NC.     | ~  | For test, do not connect (Please leave it open).                                                                                            |  |

I: Input; P: Power.

- Provide gamma voltage by AUO's FPCA, customer change to set gamma pin open.
- Modify default setting of pin "DITH".

AUO Display+

# Electrical DC Characteristics



| Item                    | Symbol           | Min.   | Typ.      | Max.     | Unit      | Remark                                |
|-------------------------|------------------|--------|-----------|----------|-----------|---------------------------------------|
| Power Voltage           | VDDIO            | 3.0    | 3.3       | 3.6      | V         | Digital Power Supply                  |
|                         | AVDD             | 10.5   | 11        | 11.5     | V         | Analog Power Supply                   |
|                         | VGH              | 14     | 15        | 16       | V         | Positive power supply for gate driver |
|                         | VGL              | -7.5   | -7        | -6.5     | V         | Negative power supply for gate driver |
| Input                   | H Level          | VIH    | 0.7xVDDIO | --       | VDDIO     | V                                     |
| Signal Voltage          | L Level          | VIL    | GND       | --       | 0.3xVDDIO | V                                     |
| Gamma reference voltage | V1 ~ V5          | AVDD/2 | --        | AVDD - 1 | V         | Note 1                                |
|                         | V6 ~ V10         | 1      | --        | AVDD/2   | V         | --                                    |
| VCOMin                  | V <sub>cdc</sub> | 3.75   | 3.95      | 4.15     | V         | Note 2                                |

Change to



| Item <sup>Ⓢ</sup>           |                      | Symbol <sup>Ⓢ</sup>           | Min. <sup>Ⓢ</sup>      | Typ. <sup>Ⓢ</sup> | Max. <sup>Ⓢ</sup>      | Unit <sup>Ⓢ</sup> | Remark <sup>Ⓢ</sup>                                |
|-----------------------------|----------------------|-------------------------------|------------------------|-------------------|------------------------|-------------------|----------------------------------------------------|
| Power Voltage <sup>Ⓢ</sup>  |                      | VDDIO <sup>Ⓢ</sup>            | 3.0 <sup>Ⓢ</sup>       | 3.3 <sup>Ⓢ</sup>  | 3.6 <sup>Ⓢ</sup>       | V <sup>Ⓢ</sup>    | Digital Power Supply <sup>Ⓢ</sup>                  |
|                             |                      | AVDD <sup>Ⓢ</sup>             | 11 <sup>Ⓢ</sup>        | 11.5 <sup>Ⓢ</sup> | 12 <sup>Ⓢ</sup>        | V <sup>Ⓢ</sup>    | Analog Power Supply <sup>Ⓢ</sup>                   |
|                             |                      | VGH <sup>Ⓢ</sup>              | 20 <sup>Ⓢ</sup>        | 21 <sup>Ⓢ</sup>   | 22 <sup>Ⓢ</sup>        | V <sup>Ⓢ</sup>    | Positive power supply for gate driver <sup>Ⓢ</sup> |
|                             |                      | VGL <sup>Ⓢ</sup>              | -7.5 <sup>Ⓢ</sup>      | -7 <sup>Ⓢ</sup>   | -6.5 <sup>Ⓢ</sup>      | V <sup>Ⓢ</sup>    | Negative power supply for gate driver <sup>Ⓢ</sup> |
| Input <sup>Ⓢ</sup>          | H Level <sup>Ⓢ</sup> | VIH <sup>Ⓢ</sup>              | 0.7xVDDIO <sup>Ⓢ</sup> | -- <sup>Ⓢ</sup>   | VDDIO <sup>Ⓢ</sup>     | V <sup>Ⓢ</sup>    | Ⓢ                                                  |
| Signal Voltage <sup>Ⓢ</sup> | L Level <sup>Ⓢ</sup> | VIL <sup>Ⓢ</sup>              | GND <sup>Ⓢ</sup>       | -- <sup>Ⓢ</sup>   | 0.3xVDDIO <sup>Ⓢ</sup> | V <sup>Ⓢ</sup>    |                                                    |
| VCOMin <sup>Ⓢ</sup>         |                      | V <sub>cdc</sub> <sup>Ⓢ</sup> | 3.85 <sup>Ⓢ</sup>      | 4.35 <sup>Ⓢ</sup> | 4.85 <sup>Ⓢ</sup>      | V <sup>Ⓢ</sup>    | Note 1 <sup>Ⓢ</sup>                                |

Note 1: Gamma suggested circuit is as follows.

For the new IC characteristics, need to change DC voltage.

→ AVDD / VGH / VCOMin

→ Update the VCOMin voltage

Cancel the gamma voltage setting from customer's system.

# Input Current

b. Current Consumption (AGND=GND=0V)

| Parameter               | Symbol | Condition    | Min. | Typ. | Max. | Unit | Remark    |
|-------------------------|--------|--------------|------|------|------|------|-----------|
| Input current for VDDIO | IVDDIO | VDDIO = 3.3V |      | 10   | 20   | mA   | Note 1, 2 |
| Input current for AVDD  | IAVDD  | AVDD = 11V   |      | 24   | 30   | mA   | Note 1, 2 |
| Input current for VGH   | IVGH   | VGH = 15V    | --   | 0.4  | 0.6  | mA   | Note 1, 2 |
| Input current for VGL   | IVGL   | VGL = -7V    | -0.6 | -0.4 | --   | mA   | Note 1, 2 |

Note 1: Test Condition is under typical Electrical DC and AC characteristics.

Note 2: Test pattern is the following picture.



Change to



b. Current Consumption (AGND=GND=0V)

| Parameter               | Symbol | Condition    | Min. | Typ. | Max. | Unit | Remark    |
|-------------------------|--------|--------------|------|------|------|------|-----------|
| Input current for VDDIO | IVDDIO | VDDIO = 3.3V |      | 9    | 15   | mA   | Note 1, 2 |
| Input current for AVDD  | IAVDD  | AVDD = 11.5V |      | 13   | 36   | mA   | Note 1, 2 |
| Input current for VGH   | IVGH   | VGH = 21V    | --   | 0.42 | 0.5  | mA   | Note 1, 2 |
| Input current for VGL   | IVGL   | VGL = -7V    | --   | 0.45 | 0.54 | mA   | Note 1, 2 |

Note 1: Test Condition is under typical Electrical DC and AC characteristics.

Note 2: Test pattern is White.

For the new IC characteristics, need to change input current.

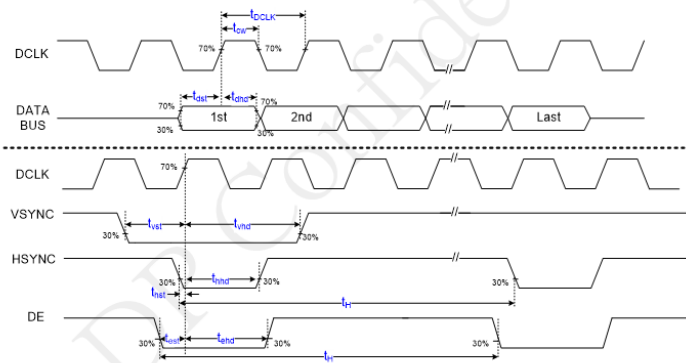
- input current for VDDIO / input current for AVDD / input current for VGH / input current for VGL
- Update the input current

# Signal AC Characteristics



## a. Signal AC Characteristics

| Parameter               | Symbol           | Min. | Typ. | Max. | Unit | Remark                                          |
|-------------------------|------------------|------|------|------|------|-------------------------------------------------|
| DCLK duty cycle         | ---              | 40   | 50   | 60   | %    | $t_{\text{low}} / t_{\text{DCLK}} \times 100\%$ |
| VSYNC setup time        | $t_{\text{vst}}$ | 0    | ---  | ---  | ns   |                                                 |
| VSYNC hold time         | $t_{\text{vhd}}$ | 2    | ---  | ---  | ns   |                                                 |
| HSYNC setup time        | $t_{\text{hst}}$ | 5    | ---  | ---  | ns   |                                                 |
| HSYNC hold time         | $t_{\text{hhd}}$ | 10   | ---  | ---  | ns   |                                                 |
| Data setup time         | $t_{\text{dst}}$ | 5    | ---  | ---  | ns   |                                                 |
| Data hold time          | $t_{\text{dhd}}$ | 10   | ---  | ---  | ns   |                                                 |
| Data enable set-up time | $t_{\text{est}}$ | 4    | ---  | ---  | ns   |                                                 |
| Data enable hold time   | $t_{\text{ehd}}$ | 2    | ---  | ---  | ns   |                                                 |

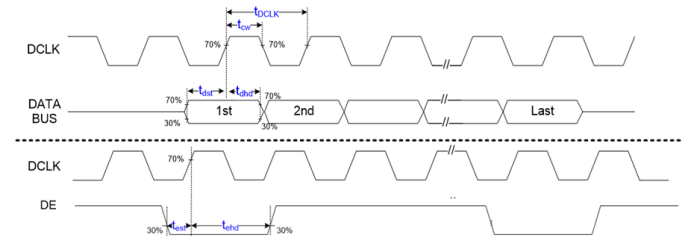


$t_{\text{H}}$ : HSYNC period  
 $t_{\text{DCLK}}$ : DCLK period  
 $t_{\text{low}}$ : the width of DCLK high



## a. Signal AC Characteristics

| Parameter       | Symbol           | Min. | Typ. | Max. | Unit | Remark                                          |
|-----------------|------------------|------|------|------|------|-------------------------------------------------|
| DCLK duty cycle | ---              | 40   | 50   | 60   | %    | $t_{\text{low}} / t_{\text{DCLK}} \times 100\%$ |
| Data setup time | $t_{\text{dst}}$ | 5    | ---  | ---  | ns   |                                                 |
| Data hold time  | $t_{\text{dhd}}$ | 5    | ---  | ---  | ns   |                                                 |
| DE setup time   | $t_{\text{est}}$ | 5    | ---  | ---  | ns   |                                                 |
| DE hold time    | $t_{\text{ehd}}$ | 5    | ---  | ---  | ns   |                                                 |



$t_{\text{DCLK}}$ : DCLK period  
 $t_{\text{low}}$ : the width of DCLK high

→ Update the signal AC Characteristics

# Timing Characteristics



Horizontal Timing<sup>1)</sup>

| Parameter <sup>1)</sup>                                                                                              | Symbol                          | Min. <sup>2)</sup> | Typ. <sup>3)</sup> | Max. <sup>4)</sup> | Unit. <sup>5)</sup> | Remark <sup>6)</sup> |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------|--------------------|--------------------|---------------------|----------------------|
| DCLK frequency <sup>1)</sup>                                                                                         | F <sub>DCLK</sub> <sup>2)</sup> | 25 <sup>3)</sup>   | 40 <sup>3)</sup>   | 45 <sup>3)</sup>   | MHz <sup>4)</sup>   | <sup>5)</sup>        |
| DCLK period <sup>1)</sup>                                                                                            | T <sub>DCLK</sub> <sup>2)</sup> | 22 <sup>3)</sup>   | 25 <sup>3)</sup>   | 40 <sup>3)</sup>   | ns <sup>4)</sup>    | <sup>5)</sup>        |
| Hsync period (= T <sub>HD</sub> + T <sub>HBL</sub> ) <sup>1)</sup>                                                   | T <sub>H</sub> <sup>2)</sup>    | 1026 <sup>3)</sup> | 1056 <sup>3)</sup> | 1183 <sup>3)</sup> | DCLK <sup>4)</sup>  | <sup>5)</sup>        |
| Active Area <sup>1)</sup>                                                                                            | T <sub>HD</sub> <sup>2)</sup>   | — <sup>3)</sup>    | 800 <sup>3)</sup>  | — <sup>3)</sup>    | DCLK <sup>4)</sup>  | <sup>5)</sup>        |
| Horizontal blanking (= T <sub>HF</sub> + T <sub>HE</sub> ) <sup>1)</sup>                                             | T <sub>HBL</sub> <sup>2)</sup>  | 226 <sup>3)</sup>  | 256 <sup>3)</sup>  | 383 <sup>3)</sup>  | DCLK <sup>4)</sup>  | <sup>5)</sup>        |
| Hsync front porch <sup>1)</sup>                                                                                      | T <sub>HF</sub> <sup>2)</sup>   | 10 <sup>3)</sup>   | 40 <sup>3)</sup>   | 167 <sup>3)</sup>  | DCLK <sup>4)</sup>  | <sup>5)</sup>        |
| Delay from Hsync to 1 <sup>st</sup> data input <sup>1)</sup><br>(= T <sub>HW</sub> + T <sub>HB</sub> ) <sup>2)</sup> | T <sub>HE</sub> <sup>2)</sup>   | <sup>3)</sup>      | 216 <sup>3)</sup>  | <sup>3)</sup>      | DCLK <sup>4)</sup>  | <sup>5)</sup>        |
| Hsync pulse width <sup>1)</sup>                                                                                      | T <sub>HW</sub> <sup>2)</sup>   | 1 <sup>3)</sup>    | 128 <sup>3)</sup>  | 136 <sup>3)</sup>  | DCLK <sup>4)</sup>  | <sup>5)</sup>        |
| Hsync back porch <sup>1)</sup>                                                                                       | T <sub>HB</sub> <sup>2)</sup>   | 80 <sup>3)</sup>   | 88 <sup>3)</sup>   | 215 <sup>3)</sup>  | DCLK <sup>4)</sup>  | <sup>5)</sup>        |

Change to



| Signal <sup>1)</sup>             | Symbol <sup>2)</sup>                | Min. <sup>3)</sup>            | Typ. <sup>4)</sup> | Max. <sup>5)</sup> | Unit. <sup>6)</sup> |
|----------------------------------|-------------------------------------|-------------------------------|--------------------|--------------------|---------------------|
| Clock Frequency <sup>1)</sup>    | 1/ T <sub>Clock</sub> <sup>2)</sup> | 32.6 <sup>3)</sup>            | 39.6 <sup>3)</sup> | 62.4 <sup>3)</sup> | MHz <sup>4)</sup>   |
| Horizontal Section <sup>1)</sup> | Period <sup>2)</sup>                | T <sub>H</sub> <sup>2)</sup>  | 890 <sup>3)</sup>  | 1000 <sup>3)</sup> | 1300 <sup>3)</sup>  |
|                                  | Active <sup>2)</sup>                | T <sub>HD</sub> <sup>2)</sup> | — <sup>3)</sup>    | 800 <sup>3)</sup>  | — <sup>3)</sup>     |
|                                  | Blanking <sup>2)</sup>              | T <sub>HB</sub> <sup>2)</sup> | 90 <sup>3)</sup>   | 200 <sup>3)</sup>  | 500 <sup>3)</sup>   |
| Vertical Section <sup>1)</sup>   | Period <sup>2)</sup>                | T <sub>V</sub> <sup>2)</sup>  | 749 <sup>3)</sup>  | 660 <sup>3)</sup>  | 800 <sup>3)</sup>   |
|                                  | Active <sup>2)</sup>                | T <sub>VD</sub> <sup>2)</sup> | — <sup>3)</sup>    | 600 <sup>3)</sup>  | — <sup>3)</sup>     |
|                                  | Blanking <sup>2)</sup>              | T <sub>VB</sub> <sup>2)</sup> | 10 <sup>3)</sup>   | 60 <sup>3)</sup>   | 200 <sup>3)</sup>   |
| Frame Rate <sup>1)</sup>         | F <sup>2)</sup>                     | <sup>3)</sup>                 | 60 <sup>3)</sup>   | <sup>3)</sup>      | Hz <sup>4)</sup>    |

Vertical Timing<sup>1)</sup>

| Parameter <sup>1)</sup>                                                | Symbol                         | Min. <sup>2)</sup> | Typ. <sup>3)</sup> | Max. <sup>4)</sup> | Unit. <sup>5)</sup> | Remark <sup>6)</sup> |
|------------------------------------------------------------------------|--------------------------------|--------------------|--------------------|--------------------|---------------------|----------------------|
| Vsync period (= T <sub>VD</sub> + T <sub>VB</sub> ) <sup>1)</sup>      | T <sub>V</sub> <sup>2)</sup>   | — <sup>3)</sup>    | 628 <sup>3)</sup>  | 635 <sup>3)</sup>  | Th <sup>4)</sup>    | <sup>5)</sup>        |
| Active lines <sup>1)</sup>                                             | T <sub>VD</sub> <sup>2)</sup>  | — <sup>3)</sup>    | 600 <sup>3)</sup>  | — <sup>3)</sup>    | Th <sup>4)</sup>    | <sup>5)</sup>        |
| Vertical blanking (= T <sub>VF</sub> + T <sub>VE</sub> ) <sup>1)</sup> | T <sub>VBL</sub> <sup>2)</sup> | — <sup>3)</sup>    | 28 <sup>3)</sup>   | 35 <sup>3)</sup>   | Th <sup>4)</sup>    | <sup>5)</sup>        |
| Vsync front porch <sup>1)</sup>                                        | T <sub>VF</sub> <sup>2)</sup>  | — <sup>3)</sup>    | 1 <sup>3)</sup>    | 8 <sup>3)</sup>    | Th <sup>4)</sup>    | <sup>5)</sup>        |
| GD start pulse delay <sup>1)</sup>                                     | T <sub>VE</sub> <sup>2)</sup>  | — <sup>3)</sup>    | 27 <sup>3)</sup>   | — <sup>3)</sup>    | Th <sup>4)</sup>    | <sup>5)</sup>        |
| Vsync pulse width <sup>1)</sup>                                        | T <sub>VW</sub> <sup>2)</sup>  | 1 <sup>3)</sup>    | 3 <sup>3)</sup>    | 16 <sup>3)</sup>   | Th <sup>4)</sup>    | <sup>5)</sup>        |
| Hsync/Vsync phase shift <sup>1)</sup>                                  | T <sub>VPO</sub> <sup>2)</sup> | 2 <sup>3)</sup>    | 320 <sup>3)</sup>  | — <sup>3)</sup>    | DCLK <sup>4)</sup>  | <sup>5)</sup>        |

→ Update the Timing Characteristics

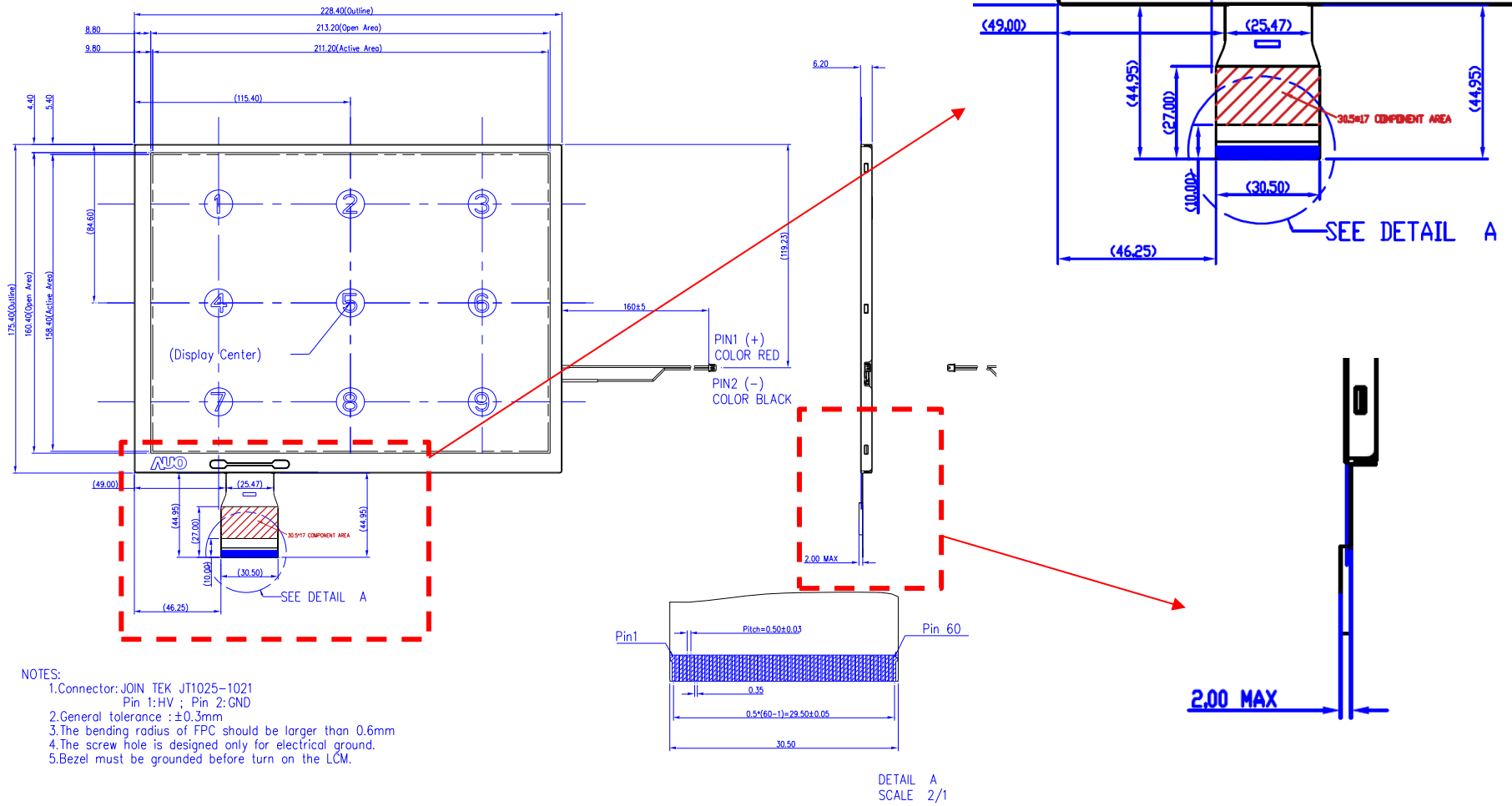




→ DE mode trigger with the rising edge of DCLK



# Outline Dimension



→ Add component on FPCA to support new IC, the FPC dimension and thickness need to modify.

AUO Display+

# Optical change

| Item         |       | Symbol | Condition        | Min.  | Typ.  | Max.  |
|--------------|-------|--------|------------------|-------|-------|-------|
| Chromaticity | White | X      | $\theta=0^\circ$ | 0.28  | 0.33  | 0.38  |
|              |       | Y      | $\theta=0^\circ$ | 0.30  | 0.35  | 0.40  |
|              | Red   | X      | $\theta=0^\circ$ | 0.550 | 0.600 | 0.650 |
|              |       | Y      | $\theta=0^\circ$ | 0.324 | 0.374 | 0.424 |
|              | Green | X      | $\theta=0^\circ$ | 0.306 | 0.356 | 0.406 |
|              |       | Y      | $\theta=0^\circ$ | 0.531 | 0.581 | 0.631 |
|              | Blue  | X      | $\theta=0^\circ$ | 0.094 | 0.144 | 0.194 |
|              |       | Y      | $\theta=0^\circ$ | 0.043 | 0.093 | 0.143 |

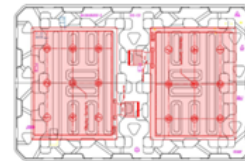


| Item         |       | Symbol | Condition        | Min.  | Typ.  | Max.  |
|--------------|-------|--------|------------------|-------|-------|-------|
| Chromaticity | White | X      | $\theta=0^\circ$ | 0.256 | 0.306 | 0.356 |
|              |       | Y      | $\theta=0^\circ$ | 0.278 | 0.328 | 0.378 |
|              | Red   | X      | $\theta=0^\circ$ | 0.543 | 0.593 | 0.643 |
|              |       | Y      | $\theta=0^\circ$ | 0.298 | 0.348 | 0.398 |
|              | Green | X      | $\theta=0^\circ$ | 0.290 | 0.340 | 0.390 |
|              |       | Y      | $\theta=0^\circ$ | 0.528 | 0.578 | 0.628 |
|              | Blue  | X      | $\theta=0^\circ$ | 0.105 | 0.155 | 0.205 |
|              |       | Y      | $\theta=0^\circ$ | 0.068 | 0.118 | 0.168 |

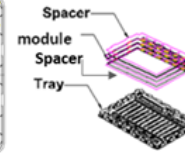
Due to new cell design, the chromaticity need to be changed.

# Packing change

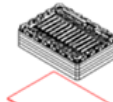
## 1. Packing Form



1 tray for 2 module + 4 spacer



19 full sets tray + 1 pcs empty tray



1 package for 19+1 trays

PP Board

PE Bag

1 package with 38pcs modules

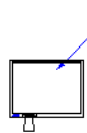
EPE Cushin

Carton

Change to



## 1. Packing Form



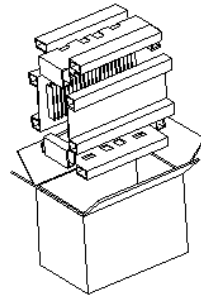
MODULE



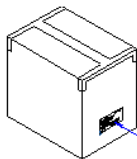
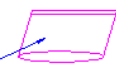
Add Protect Film and Creped Paper Tape



PUTS 20PCS MODULES in 1PCE CARTON



BAG ANTI-STATIC



CARTON LABEL

MAX. CAPACITY: 20MODULES  
MAX. WEIGHT: 11 kg  
CARTON Dim: 483(L)\*296(W)\*355(H)mm

MAX. CAPACITY : 38 MODULES  
MAX. WEIGHT : 16.159kg  
CARTON Dim : 544(L)\* 374(W)\* 376(H)mm

→ Update the new packing form

# Label change



## Carton Label

Currently



EC

**AUO Display Plus**  
MODEL NO : A104SN03 V1  
PART NO : 97D10A07.111  
AUO Corporation  
MODEL NO : A104SN03 V1  
PART NO : 97.10A07.111

QTY :

**CUSTOMER NO:**  
**CARTON NO :**

**Made In China**  
TEST  
XXX



\*ZXXX00-1942000003

## Shipping Label

Currently



EC



**AUO Display+**

# Protection Film Change

Current

Protection film attached on Polarizer



Final MP Version

Additional protection film on LCM



**Bright Innovation  Amazing Life**